

Chapter Nine Practical Projects and Conclusion (Applying Everything)

9-1: Practical Project #1: Writing a Complete Business Plan from Scratch

Goal: A 10-15 page document including executive summary, market analysis, marketing plan, financial forecast, and operational plan.

Prompting strategy for this project (combining Chapters 7 and 4):

Instead of saying "write a business plan" all at once (which gives a clichéd answer), we use the **sequential technique (7-1)** and **asking the model (7-2)**.

Opening prompt (Stage 0 — Ask the model):

*"I want to write a complete business plan for launching a **specialty boutique café with freshly roasted coffee** in a student area. The initial capital is 500 million tomans. You, as a **business consultant with 20 years of experience**, before writing the plan, ask me any questions you need to complete this plan (such as square footage, number of staff, pricing, competitors, monthly sales target). Until I fully answer all your questions, do not write anything."*

(The model asks its questions, you answer, and then you continue...)

Stage 1 (Step by step with CoT):

*"Now that you have the complete information, write the business plan **step by step** in 5 separate sections: 1) Market and competitor analysis, 2) Operational plan and equipment, 3) Marketing plan and customer acquisition, 4) 12-month financial forecast, 5) Risk analysis. Start each section with a separate heading."*

Stage 2 (Tree of Thoughts — ToT for marketing):

*"In the marketing section (Section 3), instead of one path, examine **3 different scenarios** together: Scenario A (student discounts), Scenario B (nightly coffee-tasting events), Scenario C (collaboration with office catering services). Write the pros and cons of each and ultimately suggest the best one for the student area."*

9-2: Practical Project #2: Designing a Training Course with Prompts

Goal: Designing a 10-session online course for teaching "Introductory Prompt Writing" (the very thing you are learning now!).

Prompting strategy (combining Chapter 5 personification + Chapter 8 iterative refinement):

Opening prompt:

*"As a **university professor in the field of artificial intelligence** who has taught undergraduate students for years, design a **10-session syllabus** for the course 'Fundamentals of Prompt Writing.' The audience is 20 to 40-year-olds with no technical background. **[Format]** Each session should include: title, learning objective, and one practical exercise. **[Tone]** Completely educational and fluent, like a caring teacher."*

After receiving the initial response (iterative refinement with technique 8-2):

**"The syllabus you wrote is excellent, but fix these 3 things:*

- 1. Make sessions 5 and 6 (which are about advanced techniques) simpler and dedicate an extra session to 'debugging.'*
- 2. For each session, add a 'audience profile' that says for which job (programmer, content producer, or business manager) this session is more useful.*
- 3. At the end of the course, design a final project where the student must write a comprehensive prompt for a real scenario."**

9-3: Practical Project #3: A Conversational Bot for Customer Support

Goal: Designing a system prompt for a store chatbot that answers user questions.

Prompting strategy (combining Chapter 4 conditions + Chapter 7 context management):

Main system prompt (which you give to the model at the beginning of each conversation):

*"You are a store support assistant for an online **sportswear** store. Follow these rules exactly:*

- **[Condition 1]** If the user asks about **size or sizing guide**, give the sizing chart link and ask for their height and weight to give a more precise recommendation.*
- **[Condition 2]** If the user complains about **quality or fabric**, apologize, take the order code, and say quality experts will contact them within the next 24 hours.*
- **[Condition 3]** If the user thinks **prices are expensive**, remind them of the monthly special discounts and tell them to follow the discount newsletter.*
- **[Condition 4]** If the user merely greets (hello, how are you?), respond warmly and ask, 'How can I help you find the best sports set today?'*

[Conversation management] After every 5 exchanges, if the user asks a new question, first give a 1-line summary of their previous requests to make sure you haven't mixed anything up."*

9-4: Final Checklist (Before Sending Any Prompt, Check These 7 Items)

From now on, before pressing Enter, check these 7 questions one by one like a pilot before takeoff. If you said "yes" to all of them, your prompt is almost invincible:

Row	Checklist Question	Status
1	Have I specified the Role ? (Did I tell the model who it is?)	☐ Yes / ☐ No
2	Have I given the complete Context ? (Did I state the audience, goal, constraints?)	☐ Yes / ☐ No
3	Have I started the Task with a clear operational verb? (Write, analyze, code, compare)	☐ Yes / ☐ No
4	Have I precisely determined the Format and Tone ? (Table? List? Formal? Friendly?)	☐ Yes / ☐ No
5	Have I appropriately used basic/advanced techniques (like CoT, Few-shot, or conditional)?	☐ Yes / ☐ No
6	Have I set the Constraints (Length & Level) ? (How many words? For what literacy level?)	☐ Yes / ☐ No
7	Am I ready for iterative refinement ? (Keep in mind it may not be good at first and I need to give feedback)	☐ Yes / ☐ No

Tip: Keep this checklist on a piece of paper or a notepad file next to you so that for the first 10 times you write a prompt, it's in front of your eyes. After 10 times, it becomes second nature!

9-5: Introducing Supplementary Resources and Tools (To Continue Your Professional Path)

This booklet is not the end of the road; rather, it is a beginning for entering a more professional world. These tools and resources help you always stay up to date:

 **Websites and Free Courses (For Further Reading):**

- **Learn Prompting (learnprompting.org):** The largest free prompt engineering encyclopedia in English (it also has a Persian section).
- **Prompt Engineering Guide (promptingguide.ai):** The official and scientific guide from major companies.

✂ Prompt Management and Storage Tools (Don't Lose Your Good Prompts Anymore!):

- **AIPRM (Chrome extension):** A huge library of ready-made and professional prompts that you can use with one click. (You can also save your own prompts in it).
- **Notion / Simple Excel:** Build a personal database with columns: "Prompt title," "Role," "The prompt itself," "The result I got," "Date." This way, you don't have to write a repetitive prompt each time — you just copy it!

🤖 Advanced Testing Platforms:

- **OpenAI Playground:** A professional testing environment where you can manipulate parameters like "Temperature."
- **Claude Console:** Great for testing long prompts and analyzing large documents.

🏆 Final Summary of the Booklet (From Zero to One Hundred)

My friend, let's do a quick review of all 9 chapters we went through together:

1. **Chapter 1 (Fundamentals):** You learned that a prompt is a communication bridge and the model is a word prediction machine.
 2. **Chapter 2 (Anatomy):** You became familiar with the 5 main components (role, context, task, format, tone).
 3. **Chapter 3 (Basic Techniques):** You know Zero-shot, Few-shot, magic words (step by step), and constraints.
 4. **Chapter 4 (Advanced Techniques):** You mastered Chain of Thought (CoT), Tree of Thoughts (ToT), self-consistency, and conditionals.
 5. **Chapter 5 (Personification):** You build personas, imitate writers' styles, and adjust the knowledge level.
 6. **Chapter 6 (Specialization):** You write prompts for text, code, Excel, and images.
 7. **Chapter 7 (Conversation Engineering):** You advance the conversation step by step, ask the model questions, and manage the context window.
 8. **Chapter 8 (Debugging):** You recognize the 5 mistakes, do iterative refinement, escape hallucinations, and run A/B tests.
 9. **Chapter 9 (Final Project):** Now you can design a business plan, a training course, and a support bot.
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A Special Gift for You (Final Summary Prompt):

Now that you've finished this booklet, you can ask AI to build you a "prompt writer assistant"! Try this prompt:

"You, as a prompt engineer with 10 years of experience, are going to help me turn any idea into a professional prompt. Every time I tell you a simple topic, your job is to use the 7-step checklist (role, context, task, format, tone, techniques, constraints) to write a complete and ready-to-use prompt for a chatbot, and finally ask me if I want any revisions. Ready?"